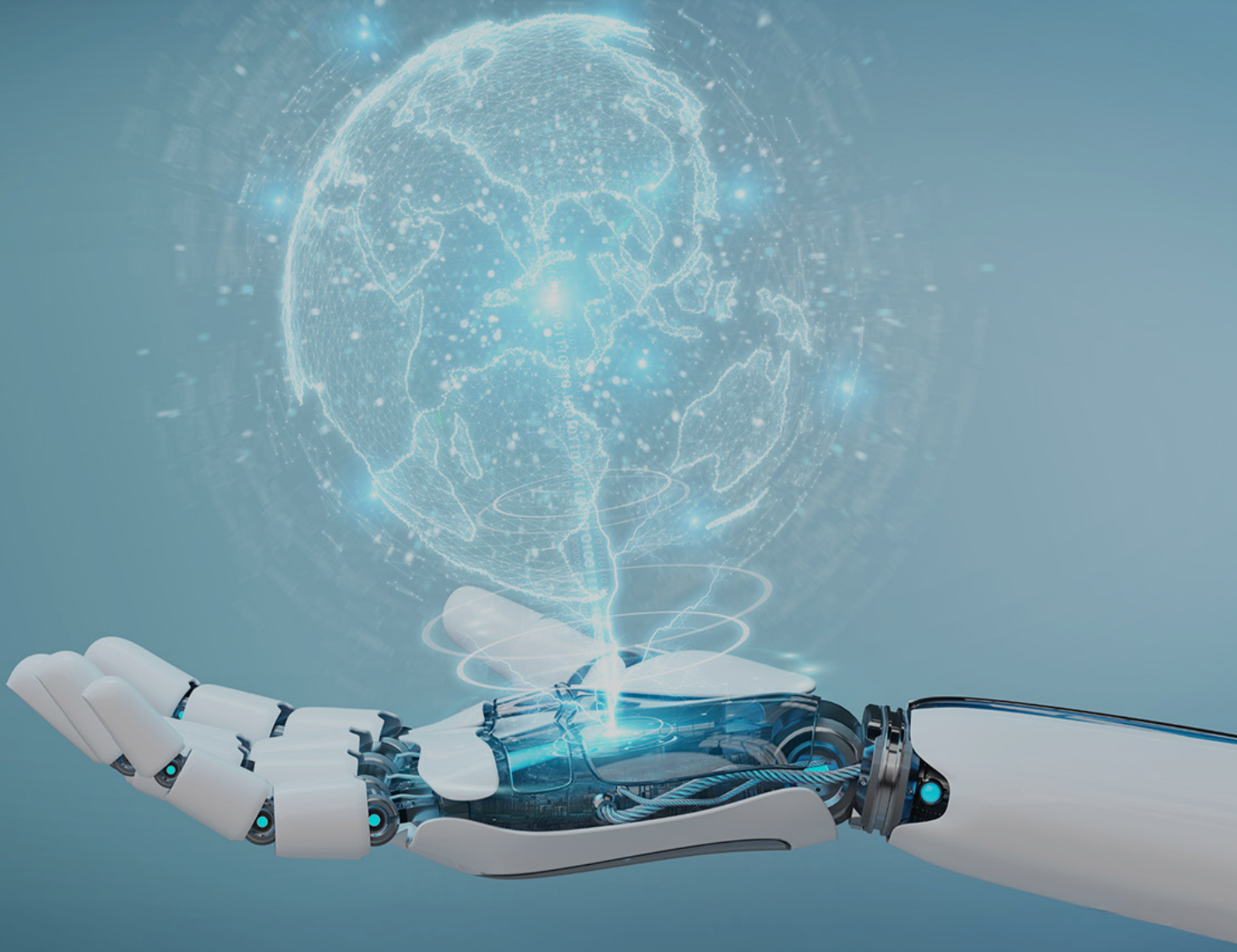
MIT Professional Education



Applied Agentic AI for Organizational Transformation

Leverage AI Agents to Elevate Efficiency and Innovate Models

“

AI is no longer just a tool—it’s a strategic imperative. Leaders who understand how to harness Generative and Agentic AI will shape the future of business, not just react to it.

Dr. Abel Sanchez

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AI Transformation: Leading with Generative and Agentic AI

In a world rapidly reshaped by artificial intelligence, strategic leadership demands more than awareness - it requires action. The MIT Professional Education course Applied Agentic AI for Organizational Transformation equips decision-makers with the insights, tools, and frameworks to lead successful AI-driven transformation across their organizations. In addition to focusing on prominent technologies, the program also dives deep into Generative and Agentic AI - empowering leaders to drive practical innovation, navigate ethical frontiers, and create measurable business value.

Designed for mid-to-senior working professionals in strategic roles, this 8-week course is your opportunity to explore the AI landscape - from foundational models and tools like ChatGPT and, to governance, compliance, and real-world adoption. Weekly hands-on mini-projects give you a safe, practical environment to prototype AI use cases.

At the end of the program, you'll emerge with an executive-ready AI adoption plan or presentation, supported by hands-on assessments and expert instruction from MIT faculty. In addition, you would be able to create a personalized AI adoption plan tailored to your organization's needs.

MIT Professional Education

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global alumni
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Why Elevate Your Digital Strategy?

$100 trillion

will be added to the world economy through digital transformation by 2025.

Source: World Economic Forum

79%

of Executives Believe AI Will Significantly Transform Their Industry in the Next 3 Years.

Source: PwC Global AI Study, 2023

76%

of executives agree that organizations need to drastically redesign the experiences that unite technology and people in more human-centered ways.

Source: Accenture

Why this program?

- Understand the building blocks of generative and agentic AI, including how they differ, how they work, and what business functions they can transform.
- Design AI-integrated digital strategies using cloud, APIs, and enterprise systems to deploy real-world solutions.
- Apply AI through mini-projects and gain hands-on experience generating images, audio, text, or video in business contexts.
- Plan for ethical considerations, legal compliance, as well as addressing risks like deepfakes, misinformation, and prompt injection with sound governance strategies.

8 weeks | 8 - 10 hours a week | Online

Learning outcomes

1. Explain the fundamentals of Generative and Agentic AI, including their architectures, use cases, and future potential.
2. Assess where and how AI can improve business outcomes, from operational workflows to customer engagement.
3. Integrate AI systems into existing digital ecosystems, leveraging cloud infrastructure, APIs, and enterprise platforms.
4. Evaluate AI platforms for strategic fit, capabilities, and risk.
5. Develop agent-based solutions to orchestrate multi-step workflows and enhance automation.
6. Navigate governance and compliance challenges, including ethical deployment and regulatory frameworks like GDPR, HIPAA, and CCPA.
7. Build a strategic AI roadmap, culminating in an actionable adoption plan or executive pitch.

Key Takeaways

01 Lead with Confidence in the Age of AI: Understand the strategic implications of Generative and Agentic AI—and how to integrate them into your business models, customer journeys, and operations.

02 Understand the Tools of Transformation: Gain hands-on exposure to industry-leading platforms such as ChatGPT, and Ollama—learning how to evaluate, prototype, and deploy AI-powered solutions.

03 Drive Responsible Innovation: Develop a clear understanding of AI ethics, bias mitigation, compliance frameworks (like GDPR and HIPAA), and governance best practices.

04 Communicate AI Value Across the Organization: Learn to articulate AI's business potential in clear, actionable language that engages stakeholders across functions.

05 Deliver a Strategic AI Roadmap: Walk away with an executive-ready AI adoption plan or strategic presentation tailored to your organization's needs.

Who is the program for?

This program is designed for decision-makers who recognize the strategic urgency of AI and want to lead its adoption effectively within their organizations. It's tailored for professionals who may not build AI tools themselves—but are responsible for choosing, funding, governing, or operationalizing them.

This program is ideal for:

C-suite executives (CEOs, CIOs, CTOs, CMOs, COOs) looking to make informed, forward-thinking decisions about AI integration.

Business leaders and functional heads driving digital innovation in areas like operations, strategy, marketing, or product.

Managers and team leads responsible for modernizing workflows and aligning teams with technology change.

Technical professionals looking to bridge the gap between engineering and leadership in digital transformation efforts.

Consultants and advisors helping clients navigate AI-powered change in complex organizational contexts.

This course assumes no prior experience with AI tools or programming. However, a strategic mindset and openness to hands-on experimentation will allow participants to get the most out of the experience.

Instructor

Dr. Abel Sanchez

Director of the MIT Master of Engineering in Advanced Manufacturing and Design. Associate Director, MIT.nano

Dr. Sanchez holds a Ph.D. from the Massachusetts Institute of Technology (MIT). He is the Executive Director of MIT's Geospatial Data Center (GDC). His areas of specialty include the Internet of Things (IOT), Big Data, Cybersecurity, and Digital Innovation. He teaches graduate courses in Data Science, Cybersecurity, and Innovation. For the past eight years his research has focused on architecting large-scale computation.

In IOT, Dr. Sanchez led the global network architecture for the Internet of Things at MIT. The design addresses large-scale computation. Compared to the largest numbers in the world, 2 billion computers, 7 billion phones, and 7 billion people, IOT is orders of magnitude bigger. In similar work, Dr. Sanchez directed simulation of the U.S. critical infrastructure with the National Infrastructure Simulation and Analysis Center (NISAC).ia.

In data science, early work included supply chain information engineering, analytics, simulation, and visualization with Wal-Mart, Kraft, and SAP. Dr. Sanchez extended his work to global anti-counterfeiting efforts with Johnson & Johnson, SAP, and Altria. Helping Altria scale track-and-trace using the Electronic Product Code Information Services standard from RFID. The work produced a global infrastructure used by industry and government to this day.

In enterprise computing, Dr. Sanchez led the design of a global data infrastructure simulator, modeling follow-the-sun engineering, to evaluate the impact of competing architectures on the performance, availability and reliability of the system for Ford Motor Company. The simulator modeled user actions, applications, background processes, network load, servers, storage, and global data centers. The work identified data center reductions opportunities estimated at a billion dollars in savings.

In cyber security, Dr. Sanchez directed impact analysis of large-scale cyber-attacks designing Cyber Ranges for the Department of Defense (DOD). Conducting repeatable experiments in impact analysis and the ability to model the cyber environment in a highly portable fashion. Looking at the insider threat Dr. Sanchez led the DOD security study on Enterprise Resource Planning systems across the United States Armed Forces.

In password security, Dr. Sanchez led the design of a password firewall (negative authentication) for the Intelligence Advanced Research Projects Activity (IARPA) agency.
The Negative Filtering or Negative Authentication (NA) approach utilizes a form of complement profiles which resembles the censoring and maturation process of T- cells in the immune system.

In machine learning, addressing financial fraud, Dr. Sanchez designed a situational awareness framework that exploits different perspectives of the same financial data and assigns risk scores to entities (e.g. payment documents) to improve false positive ratios and assist the identification of fraudulent activity in huge and unlabeled financial data in collaboration with Accenture.

In physical security, Dr. Sanchez is developing algorithms to assess risk in the integration of information technology (IT) and operations technology (OT).

Dr. Sanchez is the founder and Chief Software Architect of the Open Source RFID platform project. Dr. Sanchez' software systems are used by Samsung, NEC, NTT, Hitachi, Motorola, SAP, IBM, and Microsoft. Other software initiatives are in use by Sandia National Laboratories, MIT, and by several organizations in East Asia and Europe.

Dr. Sanchez advises companies in the Americas, Europe, Middle East, and Asia.

> "We cannot bet on the artificial intelligence, or the human being separate.
> We must get the fusion of both."

Affiliations

- MIT Geospatial Data Center (GDC)
- MIT Sociotechnical Systems Research Center (SSRC)
- MIT Institute for Data, Systems, and Society (IDSS)
- MIT Center for Complex Engineering Systems (CCES)
- MIT Center for Computational Science and Engineering (CCSE)
- MIT Consortium for Improving Critical Infrastructure Cybersecurity (IC3)
- MIT Schwarzman School of Computer Science
- MIT AutoID Laboratory

Areas of Interest and Expertise

- Machine Learning
- Cyber/Physical Security
- Enterprise Computing
- Data Science
- IOT

Courses Taught with MIT Professional Education

- Applied Generative AI for Digital Transformation
- Blockchain: Disruptive Technology
- Cloud & DevOps: Continuous Transformation
- Data Leadership: Transforming the Corporation´s Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing
- Digital Transformation: From AI and IOT to Cloud, Blockchain, and Cybersecurity

Instructor

Prof. John R. Williams

Professor of Information Engineering, Civil and Environmental Engineering and Director of MIT Geospatial Data Center, and a faculty member in the Center for Computational Science and Engineering part of the Schwarzman School of Computer Science.

John Williams holds a BA in Physics from Oxford University, a MS in Physics from UCLA, and a Ph.D. in Numerical Methods from University of Wales, Swansea. His research focuses on the application of Exascale computation to problems in cyber-physical systems, security and energy. His work on fault-tolerant computing using container migration won the IEEE, High Performance Extreme Computing Conference award for Best Innovation in 2019. He is director of MIT's Geospatial Data Center and from 2006-2012, was Director of the MIT Auto-ID Laboratory, where the Internet of Things was invented.

He is author or co-author of over 250 journal and conference papers, as well as the books on Rock Mechanics and RFID Technology.

He contributed to the 2013 report for the UK Office for Science Foresight Project- The Future of Manufacturing. Alongside Bill Gates and Larry Ellison, he was named as one of the 50 most powerful people in Computer Networks. He consults to companies including Accenture, Schlumberger, Shell, Total, Exxon, SAP Research, Microsoft Research, Kajima Corp, US Lincoln Laboratory, Sandia National Laboratories, US Intelligence Advanced Research Projects Activity, Motorola, Phillip-Morris Inc., Ford Motor Company, Exxon-Mobil, Shell, Total, and ARAMCO.

His international collaborations include Oxford and Cambridge Universities, HKUST, KACST, Alfaisal University, PolyU Hong Kong, Imperial College of Science and Technology UK, Malaysia University of Science and Technology (MUST), and Masdar Institute of Science and Technology Abu Dhabi. He organized the first Cyber-Physical Security Conference in the UK (2011) and along with Dr. Sanchez, he runs the MIT Applied Cyber Security Professional Education summer course. At MIT he teaches courses Architecting Software Systems (MIT 1.125) and Engineering Computation and Data Science (MIT 1.00/1.001).

In data engineering and data science, early work included simulation of Ford's global network, analysis of SAP smart grid billing system. For Altria, he analyzed the performance of item level tagging and also their implementation of an anti-counterfeiting system using the Electronic Product Code (EPC).

In password security, Dr. Williams was PI that developed the algorithms for a negative password authentication system for the Intelligence Advanced Research Projects Activity (IARPA) agency.

Dr. Williams advises companies in the Americas, Europe, Middle East, and Asia.

> "Once you climb that first mountain of knowledge, you will be able to teach yourself many things."

Affiliations

- MIT Department of Civil and Environmental Engineering
- MIT Center for Computational Science and Engineering (CCSE)
- MIT Schwarzman School of Computer Science
- MIT Geospatial Data Center (GDC)
- MIT Auto-ID Laboratory
- MIT Center for Complex Engineering Systems (CCES)
- MIT Consortium for Improving Critical Infrastructure Cybersecurity (IC3)
- MIT System Design and Management Program

Areas of Interest and Expertise

- Information Technology,
- Cyber/Physical Security
- Web-Based Education Technology
- Large Scale Network Simulation
- GeoNumerics of Granular and Powder Systems
- Modern Software Architecting and Cyber Security
- Web Services and Distributed Computing
- Discrete Element Simulation and Analysis of Discontinua

Courses Taught with MIT Professional Education

- Applied Generative AI for Digital Transformation
- Blockchain: Disruptive Technology
- Cloud & DevOps: Continuous Transformation
- Data Leadership: Transforming the Corporation´s Operations, Management, and Mindset to Leverage Data, AI, and Cloud Computing
- Digital Transformation: From AI and IOT to Cloud, Blockchain, and Cybersecurity

Course Outline

Module 1
Foundations of Generative and Agentic AI

1. Generative AI Fundamentals
2. AI Chatbots: Past, Present, and Future
3. Cost-Optimized Models and Performance Trade-Offs
4. Exploring Multimedia and Language Interaction Models
5. Advanced Applications of Generative AI Tools

Module 2
The Rise of Agentic AI and Emerging Platforms

1. Emerging Agentic Platforms
2. Vibe Living
3. Single vs. Multi-Agent Architectures
4. Open-Source vs. Closed-Source AI Systems

Module 3
Connecting Agents to Digital Ecosystems

1. Building Agents into Existing Workflows
2. Integrating Generative and Agentic AI with Existing Systems: Challenges and Solutions
3. Spotify MCP and Other Edge-cases of Agent Integration
4. Empathy and Response-Tuning for Customer-Facing Agents

Module 4
Agent Risks, Disinformation, and Systemic Impact

1. Classic and Current Cybersecurity Risks
2. Cybersecurity for Agent Ecosystems
3. Limitations of Agent Perception and Error Correction

Module 5
AI Agents by Business Function

1. The Maturity Cycle of Agentic Implementation (Crawl-walk-run)
2. Advantages Of Agentic AI for the Product Development Cycle (E.G. Accelerated Prototyping, Automation of Quality Assurance Pipelines, Reducing Product Launch Timelines)
3. Functional Deployments: HR Bots, Finance Advisors and IT Copilots
4. Agent Architecture in the Enterprise: Centralized Vs. Embedded

Module 6
The Last Mile – From Pilot to Practice

1. Voice Agents: Synthesis, Phone Systems, and Real-time Applications
2. Last-mile Integration: Why Pilots Succeed but Deployments Stall
3. Internal Resistance and Change Management
4. Monitoring Agent Performance (Metrics, KPIs, Feedback Loops)

Module 7
Governance, Compliance, and Agent Testing

1. Regulation Overview: GDPR, CCPA, HIPAA, and Emerging Agent Rules
2. Testing Agent Behavior: Sandboxing, A/B Testing, Safety Checks
3. Agent Speed vs. Oversight: Where to Insert Guardrails
4. Documentation and Compliance Readiness

Module 8
Strategy Capstone – Leading the Agentic Future

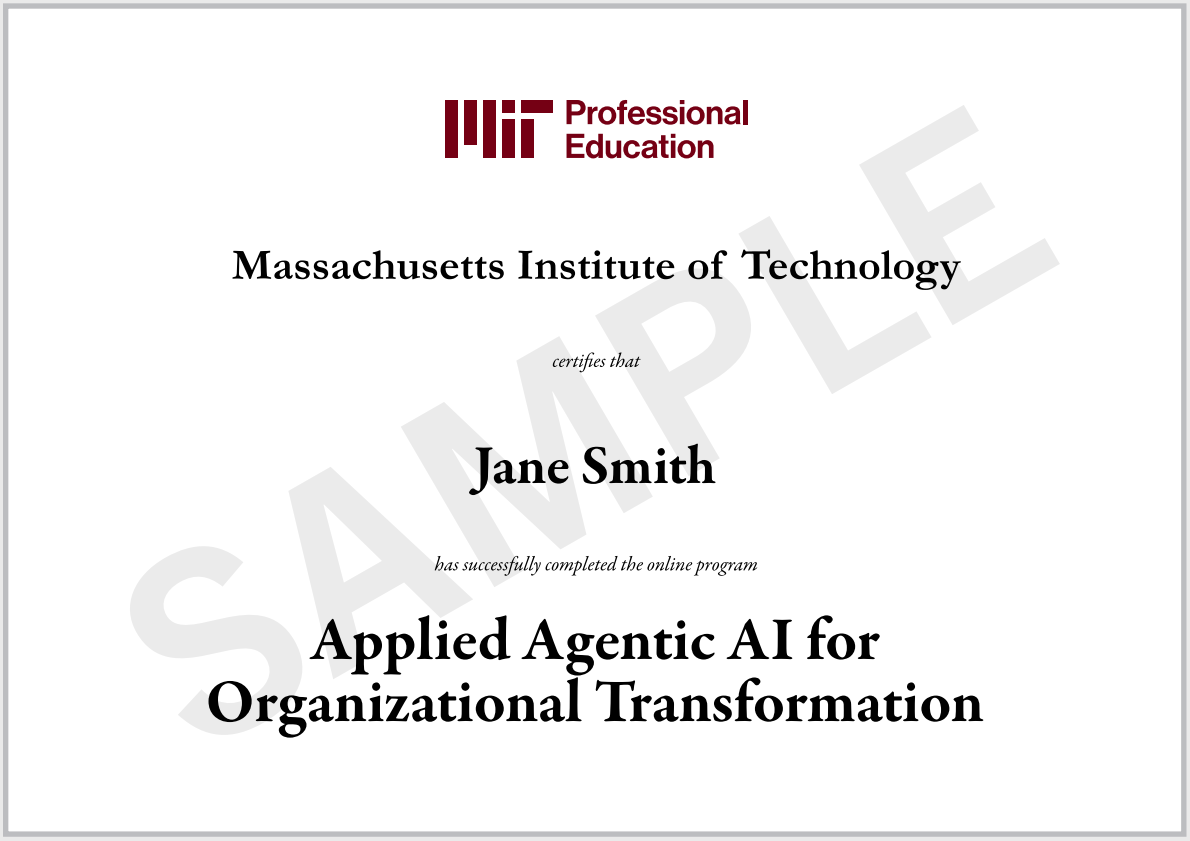
1. Strategic roadmap development (short-, medium-, long-term)
2. Team structure, vendor choice, and internal capability building
3. Organizational culture and leadership for agent adoption
4. Summary of technical enablers and business opportunities
5. Maturity assessment of agent strategy – change management

*Modules and curriculum are subject to change

Certificate of Completion

All participants who successfully complete **Applied Agentic AI for Organizational Transformation** will receive an MIT Professional Education Certificate of Completion.

In order to obtain CEUs, participants must complete a required CEU accreditation form. CEUs are calculated based on the number of learning hours in each course.



The Continuing Education Unit (CEU) is defined as 10 contact hours of ongoing learning to indicate the amount of time they have devoted to a non-credit/non-degree professional development program. To understand whether or not these CEUs may be applied toward professional certification, licensing requirements, or other required training or continuing education hours, please consult your training department or licensing authority directly.

MIT Professional Education

Welcome to the gateway

to MIT knowledge and expertise for professionals around the world.

155

Participants from 155 countries

+1600

More than 1,600 online activities

92%

92% of Professional Alumni rated their experience as outstanding

About MIT Professional Education

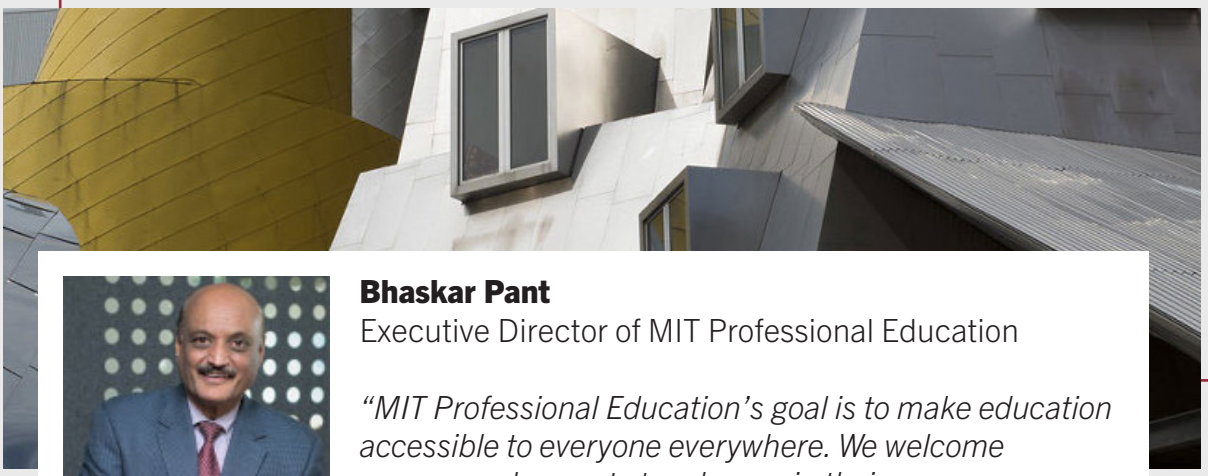
MIT Professional Education fosters the development of innovative leaders equipped to address complex problems globally

MIT Professional Education provides continuing education courses and lifelong learning opportunities for science, engineering, and technology professionals at all levels, from around the world. Worthy of note is that **MIT professors and lecturers lead and teach all MIT Professional Education offerings**. Some of our programs have a long history. Others are relatively new.

Our Digital Plus Programs go beyond online, blending innovative content delivered using the best of online technology and traditional classroom instruction in a flexible, collaborative learning environment.

MIT Professional Education is central to MIT's vision. It fulfills the mandate to connect practitioner-oriented education with industry and incorporate industry feedback and knowledge into MIT education and research.

The Institute is committed to generating, disseminating, and preserving knowledge while working with others to apply this **knowledge for the benefit of humankind.**



Bhaskar Pant
Executive Director of MIT Professional Education

"MIT Professional Education's goal is to make education accessible to everyone everywhere. We welcome everyone who wants to advance in their career or grow their business to unite and be part of the MIT Professional Education all-inclusive community."

Corporate Courses

At MIT Professional Education, we are committed to accompanying organizations as they aspire to provide their professionals with the ongoing education critical for **success in today's competitive environment**. That's why we offer plans for companies who wish to train their professionals with us.

Which program best suits your organization's needs?

1

Standard Program:
Teams or groups of at least 10 people will obtain special conditions in their registration for any of our programs.

2

Custom Programs
Custom-made to meet the specific needs of each organization and its professionals.

3

Private Cohorts:
Available in Open Enrollment courses with a minimum of participants.

For more information on our Programs for Organizations, please contact us at: **professionalprograms@mit.edu**

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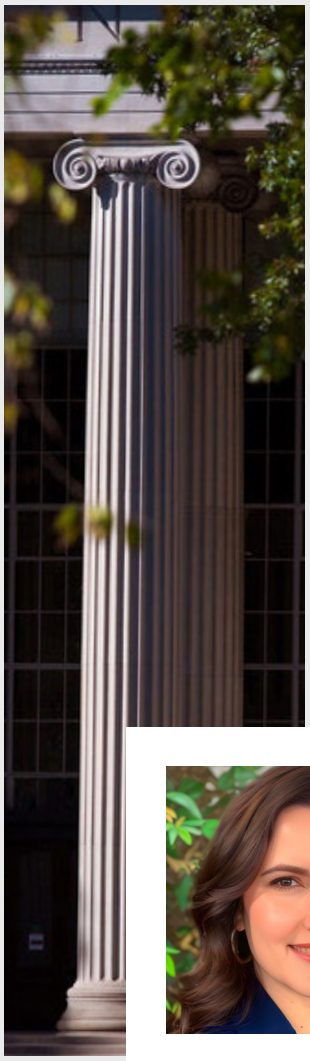
Delivered in collaboration with global alumni -excellence or nothing-

The Beyond Online Methodology

MIT Professional Education is revolutionizing the digital learning experience. In fact, we are no longer talking about online-only, but rather an interactive and collaborative learning experience that's digitally powered: a Beyond Online experience.

You will have access to one of the most innovative e-learning platforms that utilizes the latest technology.You will also have an opportunity to meet and work with some of the world's leading subject matter experts in their respective fields.



With MIT Professional Education, you'll have an opportunity to:

- Attend select live webinars with MIT Instructors.
- Collaborate and engage with world-renowned MIT Instructors.
- Learn from expert learning facilitators, guest speakers, and subject matter experts from industries.
- Weekly live webinars from learning facilitators.
- Fulfill your reskilling and upskilling needs in this fast-changing environment.
- Master the technical and human skills necessary for effective leadership.
- Meet and network with other program participants from all around the world.

Clara Piloto
Senior Director of Global Programs, Digital Plus Programs, and Corporate Programs

"We are currently amid Digital Transformation and Industry 4.0. It is now necessary to understand AI, IoT, cloud, blockchain, and cybersecurity to maintain a competitive edge professionally. Leverage the digital transformation to spark innovation and growth in your organization and in your profession."

Benefits of Joining the Professional Alumni Community

MIT Professional Education offers **a number of benefits for participants** who successfully complete our courses:

1. **15% discount** for MIT Professional Education online courses and short duration, in-person programs.
2. **Continued access to the virtual campus for six months after completion** of the course.
3. **Exclusive announcements** of new courses, programs, and events through our **virtual Professional Alumni Campus: Professional Network.**
4. Invitation to the **MIT Professional Education Professional Alumni LinkedIn group.**
5. **Networking opportunities** with other MIT Professional Education Professional Alumni.

MIT Professional Education

Are you ready to take charge of the latest technologies?

Frequently Asked Questions

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global alumni

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For more information, please contact us:

Email: **professionalprograms@mit.edu**

Website: **https://professionalprograms.mit.edu**

MIT Professional Education is collaborating with online education provider Global Alumni to offer a portfolio of online programs. These programs leverage MIT's leadership in innovation, science, engineering, and technical disciplines developed over years of research, teaching, and practice. Global Alumni collaborates with institutions to manage enrollments (including all payment services and invoicing), technology, and participant support.